BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final | Semester: Fall 2024
Duration: 2 hours | Full Marks: 50

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:

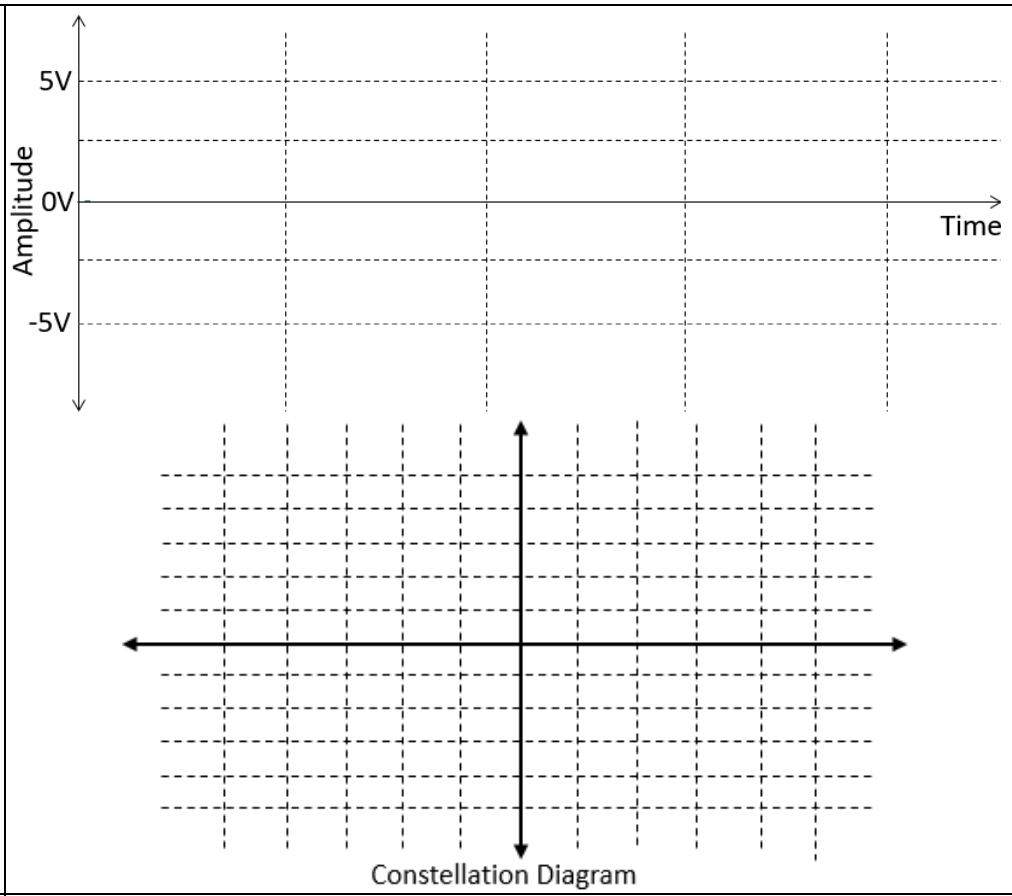
1. [CO2] a) The following table depicts a sampled analog signal for digital signal representation. By applying the concept of **Pulse Code Modulation**, assume there will be 4-bit code words for each sampled amplitude. **Show** the normalized quantized value and encoded word for the given analog signal value at different time stamps. Assume that the sampling amplitudes are between -25V to +63V. [6]

Time	Analog Signal Value (V)
0	53.3
1	-13.7
2	26.8
3	5.4

2. [CO2] a) A QPSK (Quadrature Phase Shift Keying) system transmits 2 bits per signal element with the following information: [4]

Bit Pair	Coordinates (I, Q)
00	(3,3)
01	(3, -3)
10	(-3,3)
11	(-3, -3)

The carrier signal contains 2 cycles for each signal element, and each cycle lasts 1 ms. Given the bit sequence **11100001, Draw the QPSK waveform and the constellation diagram** for an 8 ms transmission. Clearly **indicate the phase and amplitude** of the modulated waveform.

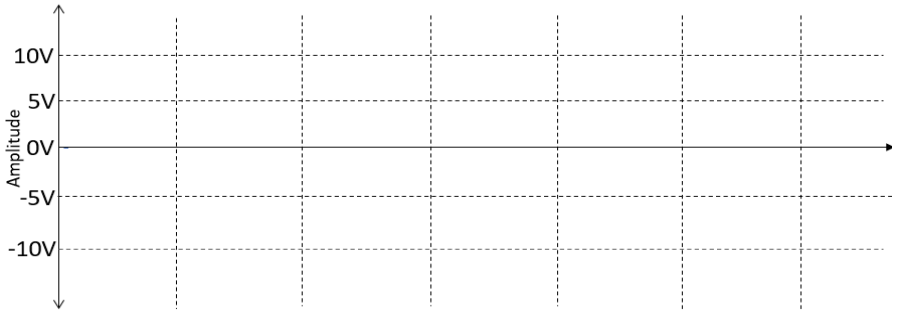


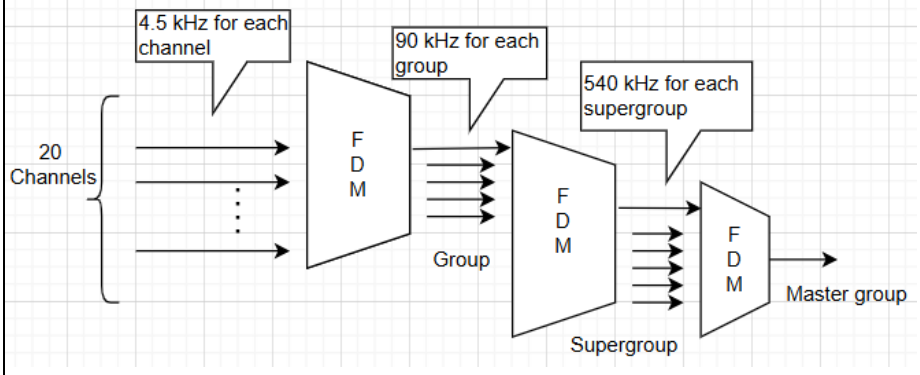
b) During another transmission, the phase sequence of a received signal was **135°**,**225°**,**45°**, and **315° respectively**. Using the above table, **decode** these phases into their corresponding bit pairs and **find** the resulting bit sequence. [2]

c) Design a FSK (Frequency Shift Keying) system to transmit the binary data stream **101101** under the following conditions: [1+3]

Each signal element has a carrier frequency **fc**=3 Hz and a frequency shift **Δf**=1Hz. The peak amplitude is **5V,** the phase is **270°**, and the duration of each signal element is **1s**.

- **Calculate** the frequencies **f1**(for binary 1) and **f0** (for binary 0).
- **Draw the FSK modulated signal** for the given binary data stream **101101**. Make sure to indicate the frequency changes for each signal element.



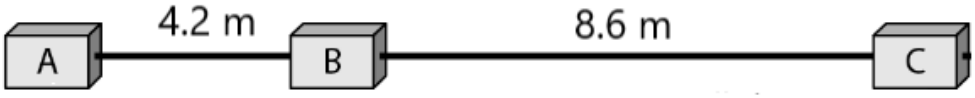
3. **[CO3]**	a)	Suppose 5 data sources with data rates of 598 Kbps, 299 Kbps, 1196 Kbps, 598 Kbps, and 299 Kbps are to be multiplexed. **Show** the visual representation of MUX input and output without adding extra pulses in the input and the number of input channels will remain the same. If a character at a time is interleaved using the concept of TDM with 2 synchronizing bits, then **answer the following questions**: • What is the frame duration? • What is the input bit duration? • What is the frame rate? • What is the output data rate? • What is the output bit duration?	[2+5]
	b)	**Analog Hierarchy:**  In the given analog hierarchy, 20 channels are multiplexed using Frequency Division Multiplexing (FDM), where each channel occupies a bandwidth of **4.5 kHz**, forming a group with a bandwidth of **90 kHz**. Five groups are further multiplexed to form a **supergroup** with a bandwidth of **540 kHz**. Finally, six supergroups are combined to form a **master group**. • Calculate the bandwidth allocated to each guard band before forming the supergroup. • Calculate the total bandwidth of the master group if the guard band between each supergroup is **220 kHz**	[4]
	c)	In which scenario would you prefer using synchronous TDM over statistical TDM? Show an example.	[3]

<table>
<tr><td>4.
[CO4]</td><td>a)</td><td>A device just received the first codeword=100101101 from the sender. The receiver’s clock cycle was 10% slower than the sender’s clock cycle so it calculated one bit less. It is given that the length of the Dataword is 5.

• Identify the divisor if the first and last bit of the divisor are 1 and the rest of the bits are 0.

• If the receiver adds 1 after the 9th bit of the codeword considering it as the missed bit, show if the receiver can detect the error or not. Justify your answer with necessary calculation.</td><td>[2+4]</td></tr>
<tr><td></td><td>b)</td><td>A block coding scheme is used to encode 4-bit Dataword into 8-bit Codeword. The following table shows the encoding scheme:

<table><tr><th>Dataword</th><th>Codeword</th></tr><tr><td>00</td><td>00001111</td></tr><tr><td>01</td><td>01100011</td></tr><tr><td>10</td><td>10010100</td></tr><tr><td>11</td><td>10111001</td></tr></table>
• Determine the number of errors this scheme can detect and correct.

• The received codeword is 10111011, which is corrupted. Determine if the original Dataword can be retrieved from it or not? If so, what is the Dataword? Show calculation.</td><td>[2+2]</td></tr>
<tr><td>5.
[CO4]</td><td>a)</td><td> 

THREE stations are sharing a 10 Mbps communication link using CSMA/CD protocol. Distance between the stations is shown in the figure. Propagation speed of the medium is 2*10^{6}m/s.
• Find the vulnerable time in this setting.

• What should be the minimum frame size in this communication link?</td><td>[3+3]</td></tr>
<tr><td></td><td>b)</td><td>Explain what happens in CSMA/CD once a collision is detected? How does CSMA reduce the collision possibility compared to ALOHA?</td><td>[2+2]</td></tr>
</table>

---END---